(a) failure kernel: the value of the first-matching declared rule (rule key A–I, below)

		Recon.	Res. dev.	Init. access	Execution	Persistence	Priv. esc.	Stealth	Def. imp.	Cred. access	Discovery	Lat. move.	C2	Collection	Exfiltration	Impact
preparation	Reconnaissance		0.7 G	0.4 B	0.05 C	0.05 C	0.05 C	0.05 C	0.05 C	0.05 C	0.05 C	0.05 C	0.05 C	0.05 C	0.05 C	0.05 C
	Resource development	0.7 G		0.3 I	0.3 I	0.3 I	0.3 I	0.3 I	0.3 I	0.3 I	0.3 I	0.3 I	0.3 I	0.3 I	0.3 I	0.3 I
intrusion	Initial access	0.9 F	0.9 F		0.02 A	0.02 A	0.02 A	0.02 A	0.02 A	0.02 A	0.02 A	0.02 A	0.02 A	0.02 A	0.02 A	0.02 A
	Execution	0.25 D	0.25 D	0.7 G		0.35 H	0.35 H	0.35 H	0.35 H	0.35 H	0.35 H	0.35 H	0.35 H	0.35 H	0.35 H	0.35 H
post-intrusion	Persistence	0.25 D	0.25 D	0.25 D	0.35 E		0.7 G	0.7 G	0.7 G	0.7 G	0.7 G	0.7 G	0.7 G	0.35 H	0.35 H	0.35 H
	Privilege escalation	0.25 D	0.25 D	0.25 D	0.35 E	0.7 G		0.7 G	0.7 G	0.7 G	0.7 G	0.7 G	0.7 G	0.35 H	0.35 H	0.35 H
	Stealth	0.25 D	0.25 D	0.25 D	0.35 E	0.7 G	0.7 G		0.7 G	0.7 G	0.7 G	0.7 G	0.7 G	0.35 H	0.35 H	0.35 H
	Defense impairment	0.25 D	0.25 D	0.25 D	0.35 E	0.7 G	0.7 G	0.7 G		0.7 G	0.7 G	0.7 G	0.7 G	0.35 H	0.35 H	0.35 H
	Credential access	0.25 D	0.25 D	0.25 D	0.35 E	0.7 G	0.7 G	0.7 G	0.7 G		0.7 G	0.7 G	0.7 G	0.35 H	0.35 H	0.35 H
	Discovery	0.25 D	0.25 D	0.25 D	0.35 E	0.7 G	0.7 G	0.7 G	0.7 G	0.7 G		0.7 G	0.7 G	0.35 H	0.35 H	0.35 H
	Lateral movement	0.25 D	0.25 D	0.25 D	0.35 E	0.7 G	0.7 G	0.7 G	0.7 G	0.7 G	0.7 G		0.7 G	0.35 H	0.35 H	0.35 H
	Command and control	0.25 D	0.25 D	0.25 D	0.35 E	0.7 G	0.7 G	0.7 G	0.7 G	0.7 G	0.7 G	0.7 G		0.35 H	0.35 H	0.35 H
objective	Collection	0.25 D	0.25 D	0.25 D	0.35 E	0.9 F	0.9 F	0.9 F	0.9 F	0.9 F	0.9 F	0.9 F	0.9 F		0.7 G	0.7 G
	Exfiltration	0.25 D	0.25 D	0.25 D	0.35 E	0.9 F	0.9 F	0.9 F	0.9 F	0.9 F	0.9 F	0.9 F	0.9 F	0.7 G		0.7 G
	Impact	0.25 D	0.25 D	0.25 D	0.35 E	0.9 F	0.9 F	0.9 F	0.9 F	0.9 F	0.9 F	0.9 F	0.9 F	0.7 G	0.7 G	

(b) lifecycle-distance kernel $d(a,b)$ over the four consensus stages, the signed stage offset $\Delta = s(b) - s(a)$ as subscript: $d = 1$ for $|\Delta| \leq 1$, $\gamma^{\Delta-1}$ forward, $\delta^{|\Delta|-1}$ backward, $d < z$ reads as 0 ($\gamma = 0.25$, $\delta = 0.25$, $z = 0.1$)

		Recon.	Res. dev.	Init. access	Execution	Persistence	Priv. esc.	Stealth	Def. imp.	Cred. access	Discovery	Lat. move.	C2	Collection	Exfiltration	Impact
preparation	Reconnaissance		1_0	1_{+1}	1_{+1}	0.25_{+2}	0.25_{+2}	0.25_{+2}	0.25_{+2}	0.25_{+2}	0.25_{+2}	0.25_{+2}	0.25_{+2}	0_{+3}	0_{+3}	0_{+3}
	Resource development	1_0		1_{+1}	1_{+1}	0.25_{+2}	0.25_{+2}	0.25_{+2}	0.25_{+2}	0.25_{+2}	0.25_{+2}	0.25_{+2}	0.25_{+2}	0_{+3}	0_{+3}	0_{+3}
intrusion	Initial access	1_{-1}	1_{-1}		1_0	1_{+1}	1_{+1}	1_{+1}	1_{+1}	1_{+1}	1_{+1}	1_{+1}	1_{+1}	0.25_{+2}	0.25_{+2}	0.25_{+2}
	Execution	1_{-1}	1_{-1}	1_0		1_{+1}	1_{+1}	1_{+1}	1_{+1}	1_{+1}	1_{+1}	1_{+1}	1_{+1}	0.25_{+2}	0.25_{+2}	0.25_{+2}
post-intrusion	Persistence	0.25_{-2}	0.25_{-2}	1_{-1}	1_{-1}		1_0	1_0	1_0	1_0	1_0	1_0	1_0	1_{+1}	1_{+1}	1_{+1}
	Privilege escalation	0.25_{-2}	0.25_{-2}	1_{-1}	1_{-1}	1_0		1_0	1_0	1_0	1_0	1_0	1_0	1_{+1}	1_{+1}	1_{+1}
	Stealth	0.25_{-2}	0.25_{-2}	1_{-1}	1_{-1}	1_0	1_0		1_0	1_0	1_0	1_0	1_0	1_{+1}	1_{+1}	1_{+1}
	Defense impairment	0.25_{-2}	0.25_{-2}	1_{-1}	1_{-1}	1_0	1_0	1_0		1_0	1_0	1_0	1_0	1_{+1}	1_{+1}	1_{+1}
	Credential access	0.25_{-2}	0.25_{-2}	1_{-1}	1_{-1}	1_0	1_0	1_0	1_0		1_0	1_0	1_0	1_{+1}	1_{+1}	1_{+1}
	Discovery	0.25_{-2}	0.25_{-2}	1_{-1}	1_{-1}	1_0	1_0	1_0	1_0	1_0		1_0	1_0	1_{+1}	1_{+1}	1_{+1}
	Lateral movement	0.25_{-2}	0.25_{-2}	1_{-1}	1_{-1}	1_0	1_0	1_0	1_0	1_0	1_0		1_0	1_{+1}	1_{+1}	1_{+1}
	Command and control	0.25_{-2}	0.25_{-2}	1_{-1}	1_{-1}	1_0	1_0	1_0	1_0	1_0	1_0	1_0		1_{+1}	1_{+1}	1_{+1}
objective	Collection	0_{-3}	0_{-3}	0.25_{-2}	0.25_{-2}	1_{-1}	1_{-1}	1_{-1}	1_{-1}	1_{-1}	1_{-1}	1_{-1}	1_{-1}		1_0	1_0
	Exfiltration	0_{-3}	0_{-3}	0.25_{-2}	0.25_{-2}	1_{-1}	1_{-1}	1_{-1}	1_{-1}	1_{-1}	1_{-1}	1_{-1}	1_{-1}	1_0		1_0
	Impact	0_{-3}	0_{-3}	0.25_{-2}	0.25_{-2}	1_{-1}	1_{-1}	1_{-1}	1_{-1}	1_{-1}	1_{-1}	1_{-1}	1_{-1}	1_0	1_0	

(c) the failure weight set = (a) × (b), as committed (`v3_persistent_backward`)

		Recon.	Res. dev.	Init. access	Execution	Persistence	Priv. esc.	Stealth	Def. imp.	Cred. access	Discovery	Lat. move.	C2	Collection	Exfiltration	Impact
preparation	Reconnaissance		0.7	0.4	0.05	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125	0.0125	0	0	0
	Resource development	0.7		0.3	0.3	0.075	0.075	0.075	0.075	0.075	0.075	0.075	0.075	0	0	0
intrusion	Initial access	0.9	0.9		0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.005	0.005	0.005
	Execution	0.25	0.25	0.7		0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.0875	0.0875	0.0875
post-intrusion	Persistence	0.0625	0.0625	0.25	0.35		0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.35	0.35	0.35
	Privilege escalation	0.0625	0.0625	0.25	0.35	0.7		0.7	0.7	0.7	0.7	0.7	0.7	0.35	0.35	0.35
	Stealth	0.0625	0.0625	0.25	0.35	0.7	0.7		0.7	0.7	0.7	0.7	0.7	0.35	0.35	0.35
	Defense impairment	0.0625	0.0625	0.25	0.35	0.7	0.7	0.7		0.7	0.7	0.7	0.7	0.35	0.35	0.35
	Credential access	0.0625	0.0625	0.25	0.35	0.7	0.7	0.7	0.7		0.7	0.7	0.7	0.35	0.35	0.35
	Discovery	0.0625	0.0625	0.25	0.35	0.7	0.7	0.7	0.7	0.7		0.7	0.7	0.35	0.35	0.35
	Lateral movement	0.0625	0.0625	0.25	0.35	0.7	0.7	0.7	0.7	0.7	0.7		0.7	0.35	0.35	0.35
	Command and control	0.0625	0.0625	0.25	0.35	0.7	0.7	0.7	0.7	0.7	0.7	0.7		0.35	0.35	0.35
objective	Collection	0	0	0.0625	0.0875	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9		0.7	0.7
	Exfiltration	0	0	0.0625	0.0875	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.7		0.7
	Impact	0	0	0.0625	0.0875	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.7	0.7	

rule key (match order; value × d gives the cell in (c)):

A `ia_gate_foothold` = 0.02
B `recon_gate_initial_access` = 0.4
C `recon_gate_deep` = 0.05
D `preintrusion_damper` = 0.25
E `execution_damper` = 0.35
F `backward` = 0.9
G `lateral` = 0.7
H `forward_from_foothold` = 0.35
I `forward_from_prep` = 0.3